

Education

University of Toronto

September 2024 - December 2025 (Expected)

MSc in Applied Computing (MScAC), Applied Mathematics stream.
2024 Vector Scholar in Artificial Intelligence (\$17,500)
Coursework: Generative NN, Nonlinear Numerical Analysis, Physics-Based Animation, Physics-Informed Neural Representations for Visual Computing, Communication for Computer Scientists, Technical Entrepreneurship.

University of Toronto

September 2020 - June 2024

HBSc Mathematics and Physics Specialist. High distinction.
Coursework: Machine learning, Deep learning, numerical analysis, optimization, real analysis, complex analysis, PDEs, linear algebra, Fourier analysis, probability, topology, differential geometry, nonlinear physics, quantum information theory.

Experience

1. Deep Learning Researcher · Vector Institute

November 2023 - Present

- Conduct extensive empirical experimentation and theoretical exploration on the transformer architecture.
 - Discovered novel phenomenon coined Transformer Block Coupling, displayed positive correlation with generalization.
 - Performed extensive experimentation on hundreds of open source transformer-based models.
 - Submitted work to ICML, NeurIPS and ICLR. See a [preprint](#) here.
2. Algorithms Researcher · University of Toronto, Department of Mathematics

April 2023 - Present

- Developing algorithms for solving scalar differential equations and for numerical integration.
 - Our methods dominate over alternative methods in the high-frequency regime.
 - Independently decreased time complexity by a factor of n by exploiting an over-determined system.
 - Proved algorithm correctness by establishing theoretical bounds on computation error.
 - Establish theoretical framework, implement, test and iteratively optimize, publish results in academic journals.
3. Teaching Assistant · University of Toronto

September 2023 - Present

- Run tutorials, host office hours, mark and invigilating exams.
 - Fall 2024: [MAT292](#), ODEs for second-year engineering students.
 - Winter 2024: [MAT187](#), Calculus 2 for first-year engineering students.
 - Fall 2023: [MAT244](#), ODEs for second-year math major students.
4. Machine Learning Researcher · DUNE × University of Toronto

September 2023 - April 2024

- Trained a graph neural network to detect tau neutrinos and their decay products.
 - Combatted an imbalanced dataset by implementing a modified, weighted loss function.
 - Improved classification performance on underrepresented training samples by over 20%.
5. Tutor · Forest Hill Tutoring

April 2023 - August 2024

- High school math and physics tutoring, largely focused on helping grade 11 and 12 IB students.
 - Principal instructor for Ontario credited courses such as MDM4U, MHF4U and MCV4U.
6. Structural Engineering Assistant · Safeline Construction Group

April 2022 - October 2022

- Worked on a 57 storey building in downtown Toronto assisting the senior and assistant supervisors.
 - Independently managed interior renovation of a small townhouse in downtown Toronto.
 - Worked alongside mechanical, civil, and electrical engineers throughout entire construction process.

Awards, Publications & Talks

Vector Scholarship in Artificial Intelligence. Awarded by the Vector Institute. Masters entrance scholarship valued at \$17,500.	2024
Transformer Block Coupling and its Correlation with Generalization in LLMs. <i>Preprint.</i> Murdock Aubry, Hoaming Meng, Anton Sugolov, Vardan Papyan. Work affiliated with experience (1). PDF.	2024
A Solver for Linear Scalar ODEs Whose Running Time is Bounded Independent of Frequency. <i>Preprint.</i> Murdock Aubry, James Bremer. Work affiliated with experience (2). PDF.	2023
On the Two Dimensional Levin Method. <i>Preprint.</i> Murdock Aubry, Kirill Serkh, James Bremer. Work affiliated with experience (2). PDF.	2024
The Emergence of Clusters in Self-attention Dynamics. <i>Talk.</i> UofT Graduate Applied² Math Seminar. Work affiliated with experience (1). Slides.	2023
Development of Graph Neural Networks for Tau (τ) Neutrino Detection. <i>Talk.</i> CERN and Deep Underground Neutrino Experiment (DUNE). Work affiliated with experience (4). Slides.	2023
On the Adaptive Levin Method. <i>Talk.</i> Canadian Undergraduate Mathematics Conference. Work affiliated with experience (2). Slides.	2023

Technical Skills

- Languages.** Python, MATLAB, Fortran, HTML, CSS, SQL, Bootstrap, Bash, Mathematica, $\text{\LaTeX}$, Markdown.
 - Packages.** PyTorch, Transformers, TensorFlow, Numpy, Scipy, CUDA, Pandas, Matplotlib, Seaborn, JSON, OS, Selenium, BeautifulSoup, ezdxf, dxfgrabber.
 - Software.** Git, SSH, SLURM, Tensorboard, AutoCAD, Unix.